Find the Mistake: Signs and Denominators in Conic Equations
Answer Key

Algebra 2

Quick Answers

1. (a) $= -5$, —
2. 7
3. (a) $= 5$, —
4. 12
5. 5
6. (a) the centre x $= -3$, (a) the centre y $= 4$, (b) $= 6$, —
7. $\frac{3}{4}$
8. (a) $= 3$, (b) $= -5$, —

What is verified

11 of 15 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 1, 3, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2
HSG-GPE.A.1–A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

1. *If they got 5: read the number inside the bracket as printed instead of flipping its sign*
2. *If they got 49: reported the right-hand side as the radius instead of taking its square root*
3. *If they got 3: assumed the denominator under the x term is always the larger one, without comparing the two*
4. *If they got 13.93: added the two denominators instead of subtracting, applying the hyperbola rule to an ellipse*
5. *If they got 2.65: subtracted the denominators as for an ellipse instead of adding*
6. *If they got 6: copied the linear coefficient straight across as the centre coordinate, without halving it or flipping its sign*
6. *If they got −8: copied the linear coefficient straight across as the centre coordinate*
7. *If they got 1.3333: inverted the ratio, putting the denominator of the negative term on top*
8. *If they got 4.47: took the square root of the constant term as the radius, without completing either square*
8. *If they got 5: halved the linear coefficient correctly but did not reverse its sign*

1. $(x+5)^2+(y-2)^2=36$; a student gave the centre as $(5,-2)$. (a) Correct x-coordinate. (b) Why negative.

Solution (a): The standard form is $(x-h)^2$, so $(x+5)^2$ has to be read as $\big(x-(-5)\big)^2$.

$\boxed{-5}$

(The y-coordinate the student gave is wrong for the same reason, in the other direction: $(y-2)^2$ gives $k=2$, not -2.)

Solution (b) — instructor-judged. The subtraction is built into the standard form, so a plus sign inside a bracket can only come from subtracting a negative number. Reading the printed digits straight off gives the centre reflected through the origin. Full credit refers to the subtraction in the standard form; "you flip the sign" with no reason attached is partial credit.

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2. $(x-1)^2+(y+4)^2=49$; a student gave the radius as 49.

Solution: The right-hand side of the standard form is r^2, not r.

$$r=\sqrt{49}$$

$\boxed{7}$

A quick sanity check: a radius of 49 would put the circle's edge nearly 50 units from a centre only 1 unit from the origin — far larger than the equation's numbers suggest.

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3. $\frac{x^2}{9}+\frac{y^2}{25}=1$; a student gave the semi-major axis as 3. (a) Correct value. (b) The error.

Solution (a): Compare the denominators: $25>9$, so the larger one, under y^2, carries the major axis.

$$\sqrt{25}$$

$\boxed{5}$

Solution (b) — instructor-judged. The student assumed the x denominator is always the larger one and took $\sqrt{9}=3$. The major axis belongs to whichever denominator is bigger, and here that is the y term, so the major axis runs *vertically*, from $(0,-5)$ to $(0,5)$. Full credit states the compare-the-denominators rule and says the axis is vertical. Reporting only the corrected number is not a diagnosis.

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4. $\frac{x^2}{169}+\frac{y^2}{25}=1$; a student computed $\sqrt{169+25}\approx 13.93$.

Solution: This is an ellipse, so the foci lie *inside* it and the rule subtracts.

$$c=\sqrt{169-25}=\sqrt{144}$$

$\boxed{12}$

The student used the hyperbola rule. The check that catches it every time: a focus of an ellipse must be closer to the centre than the vertex, and 13.93 is further out than the vertex at 13.

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5. $\frac{x^2}{16}-\frac{y^2}{9}=1$; a student computed $\sqrt{16-9}\approx 2.65$.

Solution: This is a hyperbola, so the rule adds.

$$c = \sqrt{16+9} = \sqrt{25}$$

$$\boxed{5}$$

The mirror image of problem 4's error. Here the check is the other way round: a hyperbola's focus must lie *beyond* its vertex, and 2.65 is inside the vertex at 4.

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6. $x^2+y^2+6x-8y-11=0$; a student gave the centre as $(6,-8)$. (a) Centre. (b) Radius. (c) The two mistakes.

Solution (a): Complete both squares. Half of 6 is 3 and half of -8 is -4:

$$(x^2+6x+9)+(y^2-8y+16) = 11+9+16$$
$$(x+3)^2+(y-4)^2 = 36$$

Reading $(x-h)^2+(y-k)^2$ gives $\boxed{x=-3}$

and $\boxed{y=4}$

.

Solution (b): The right-hand side is $r^2 = 36$, so the radius is $\sqrt{36}$. $\boxed{6}$

Solution (c) — instructor-judged. Two separate faults, and both must be named. First, the linear coefficients were copied across without being *halved*: completing the square puts half of each coefficient inside its bracket. Second, the signs were not *reversed*: the standard form subtracts, so $+3$ inside the bracket means a centre coordinate of -3. Full credit names the halving and the sign reversal as two distinct moves, since the quoted answer got both wrong.

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7. $\frac{y^2}{36}-\frac{x^2}{64}=1$; a student gave the asymptote slopes as $\pm\frac{8}{6}$.

Solution: The positive term is the one in y, so its denominator supplies the numerator of the slope: the asymptotes are $y=\pm\frac{a}{b}x$ with $a=\sqrt{36}=6$ and $b=\sqrt{64}=8$.

$$\frac{6}{8}$$

$$\boxed{\frac{3}{4}}$$

The student used $\frac{8}{6}$, the reciprocal — the slope pattern for a left-right hyperbola, applied to an up-down one. A sketch settles it: this hyperbola's branches open vertically but its asymptotes still rise less steeply than 1, because $6<8$.

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8. $x^2+y^2-4x+10y+20=0$; one student said $r=\sqrt{20}$, another said the centre is $(2,5)$.

Solution (a): Complete both squares. Half of -4 is -2 (adding 4); half of 10 is 5 (adding 25):

$$(x^2-4x+4)+(y^2+10y+25) = -20+4+25$$
$$(x-2)^2+(y+5)^2 = 9$$

The radius is $\sqrt{9}$. $\boxed{3}$

Solution (b): From $(y+5)^2$, the y-coordinate of the centre is $\boxed{-5}$

Solution (c) — instructor-judged. The two errors are different and both must be named. The first student treated the constant term as if it were already r^2 and took $\sqrt{20}$; in fact the constant has to move across and the two completing constants (4 and 25) have to be added, which turns -20 into 9. The second student halved the linear coefficient correctly but kept its sign: $+10$ halves to 5, and the standard form then subtracts, giving -5. Full credit distinguishes the two faults. Describing one error twice, or calling both "careless arithmetic", does not.